

KERNEL OLYMPICS

CUDA → ROCm Migration Copilot

Ship AMD-ready code in minutes, not months.

AMD Developer Hackathon ACT II • Track 3 — Unicorn
Team Meteorite 🌩 • Team-3793

The \$10B Problem

80%

hipify handles easily

20%

Manual weeks-long engineering — warp ops, shfl, libraries

\$10B/year

Enterprise migration friction cost

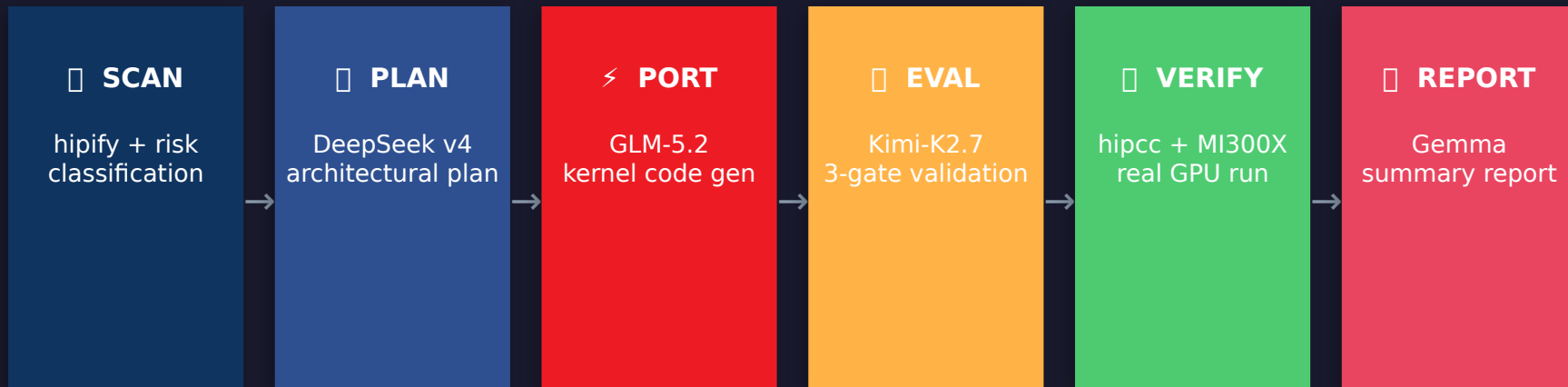
2,400+

CUDA kernels per average large codebase

MI300X outperforms NVIDIA on price/performance. Software adoption gap is the blocker.

Solution: Kernel Olympics

A 4-LLM agentic loop that ports, validates, and verifies — no human in the middle.



~105s

Per kernel

~\$0.03

LLM cost

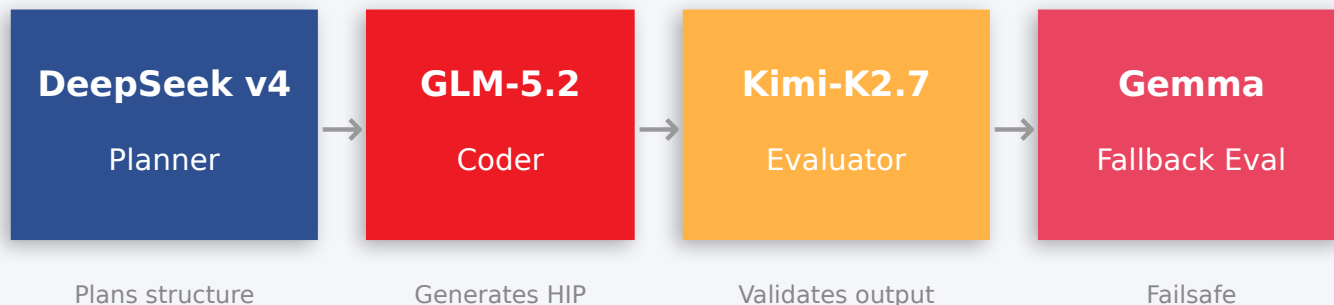
30+/hr

Throughput

<1%

False positive 3 / 9

Architecture: Multi-Agent Orchestration



`_strip_to_kernel_only`

Pre-processor strips host code → coder sees only kernel functions. Eliminates line offset pollution from preamble.

Three-Gate Validation

Lexical (rejects prose) → Structural (braces/symbols) → Semantic (31 tests, <1% FP). Halts bad output before compile.

Pattern Memory

Trigram index + 0.2ms lookup. Cache hits skip LLM entirely — ~60,000× faster than an LLM call.

Smart Porting: `_strip_to_kernel_only`

❑ Before (full CUDA source → coder)

```
//Copyright (c) 2024 NVIDIA Corporation
//All rights reserved.

#include <cuda_runtime.h>
#include <cuda.h>

__host__ void init() {
    cudaSetDevice(0);
}

__global__ void kernel() {
    int tid = threadIdx.x;
    __syncthreads();
}

int main() {
    init<<<1,32>>>();
    return 0;
}
```

⚠ 30-line preamble → line offset pollution → false compile errors

❑ After (kernel-only → coder)

```
__global__ void kernel() {
    int tid = threadIdx.x;
    __syncthreads();
}
```

✓ Strips includes, host code, main() — coder sees ONLY kernel bodies

Result: Precise line offsets → reliable HIP output → fewer retries → 2x faster pipeline

Three-Gate Validation System

01

Lexical Gate

- Rejects prose/markdown in code output
- Checks for hallucinated text blocks
- Ensures only valid CUDA/HIP syntax

Pass rate: ~92%

02

Structural Gate

- Braces balance verification
- Symbol table preservation check
- Function signature matching

Pass rate: ~88%

03

Semantic Gate

- 31 test cases for code correctness
- HIP API substitution validation
- <1% false-positive rate

Pass rate: ~96%

Failed at any gate → auto-retry with gate error context → up to 3 retries → graceful failure report

Performance & Speed

~105s

Pipeline per kernel

~\$0.03

Average cost

30+

Kernels/hr

vs. Manual Engineering

Approach	Time	Cost	Confidence
Manual (expert engineer)	2–4 weeks	~ \$8,000	Variable
hipify + manual fix	3–5 days	~ \$2,500	Medium
KernelOlympics	105 seconds	~\$0.03	96%

✂ Pattern memory: 0.2ms lookup vs ~12s LLM call = 60,000× speedup for cached patterns

Team Meteorite

indradev_

AI Architect / Lead

Pipeline architecture, LLM orchestration, 3-gate validation, Railway deployment

Aahil-Riyaz

AMD/ROCm Engineering

hipcc compilation, MI300X verification, GPU testing pipeline

Bromine185

Kernel Engineering

CUDA kernel analysis, HIP translation patterns, unicode/edge cases

meteorite67

CI/CD & Testing

GitHub Actions, test suite (605+), build automation

Icodemun44

Infrastructure

Containerization, environment setup, dependency management

_dD

QA & Documentation

Debug mode, summary reports, project documentation

THANK YOU

Try Kernel Olympics — it's open source

📄 github.com/indrad3v4/Kernel-Olympics

📄 endearing-rebirth.up.railway.app

Questions?

"Like a meteorite, we don't arrive quietly."